

Exercise sheet N°1 (Part 1: Logic, Proof, and Sets)

Exercise 1

Let P, Q and R be three statements.

0. Knowing that $(P \implies Q) \iff (\bar{P} \vee Q)$, give the truth table of $P \iff Q$.
1. Knowing that P and Q are trues and R is false, determine the truth value of $(P \vee (Q \wedge R)) \wedge (Q \vee R)$.
2. Determine the truth values of the following statements:
 $A : P \wedge (P \vee Q) \iff P,$ $B : (P \wedge \bar{P}) \implies Q,$
 $C : (\bar{P} \wedge \bar{Q}) \implies (P \implies Q),$ $D : (P \implies Q) \vee (Q \implies P),$
 $E : (P \wedge Q) \vee R,$ $F : P \wedge (Q \vee R).$

Exercise 2

For each of the following statements, indicate if it is true or false, then give its negation:

- (1) $\forall x \in \mathbb{R}, x^2 + 1 \geq 2.$ (2) $\forall m \in \mathbb{Z}, (m^2 - 1) \in \mathbb{N}.$
- (3) $\exists m \in \mathbb{Z}, \frac{m}{2} \in \mathbb{Z}.$ (4) $\exists x \in \mathbb{R}, x^2 + 1 \geq 2.$
- (5) $\exists a \in \mathbb{R}, \forall x \in \mathbb{R}, 2a + ax = x + 2.$ (6) $\forall \varepsilon > 0, \exists \alpha > 0, |x| < \alpha \implies |x^2| < \varepsilon.$

Exercise 3

Use the quantifiers $\forall$ and $\exists$ to express the following statements:

1. The square of every real number is positive.
2. Some real numbers are greater than their squares.
3. The product of two odd integers is odd.
4. The real function f can equal 0 on $\mathbb{R}$.
5. An integer is even if and only if its square is even.
6. The product of two even integers is even.

Exercise 4 *

Using the indicated proof method, prove the following statements.

1. Direct proof
 - (1) $\forall n \in \mathbb{N}, n \text{ is odd} \implies (n^2 - 1) \text{ is even}.$
 - (2) $\forall n \in \mathbb{N}, n \text{ is even} \implies n^2 \text{ is even}.$
2. Proof by contrapositive
 - (3) $\forall n \in \mathbb{N}, n^2 \text{ is even} \implies n \text{ is even}.$
 - (4) $(\forall \varepsilon > 0, |x| \leq \varepsilon) \implies x = 0$
3. Proof by contradiction
 - (5) $\forall n \in \mathbb{N}, n^2 \text{ is even} \implies n \text{ is even}.$
 - (6) $\sqrt{2} \notin \mathbb{Q}$
4. Proof by induction (**homework**)
 - (7) $\forall n \geq 5, n! \geq 2^n$
 - (8) $\forall n \in \mathbb{N}, 2^n > n$
5. Proof by counter-example
 - (9) $\forall x, y \in \mathbb{R}, x + y \leq 0$ is false.
 - (10) $\forall x \geq 0, \forall y \geq 0, x^2 + y^2 > 0$ is false.

Exercise 5

1. Let E and F be two sets defined by: $E = \{\text{All integers between } -\sqrt{2} \text{ and } \pi\}, F = \{1, 2, 3\}.$ Define E by extension and define F as comprehension.

2. We consider the following sets:

$$A = \{1, 2, 3, a, b\}, \quad B = \{a, 1, 4\}, \quad D = \{a, b, 1, 4\}$$

2.1 Can we write $1 \subset A, \{1\} \subset A, 1 \in A, \emptyset \in A, \emptyset \subset A, \{\emptyset\} \subset A?$

2.2 Determine: $A \cap B, A \cup B, A \setminus B, B \setminus A, A \triangle B, A \times B, B \times A,$ and $C_B^D.$